AI & Machine Learning in RE/MAX Smile AI

Concise technical report — models, training approach, application, and limitations

1. Project approach

The project combines machine-learning-ready pipelines with deterministic business rules. This hybrid architecture keeps the platform operational when training data is insufficient: an ML artifact is used when available, otherwise a transparent fallback produces a safe result.

Current deployment status: the repository contains no saved *.joblib* training artifacts. Therefore, the actively deployed learning component is the runtime content-based KNN agent–property matcher. Lead conversion, price estimation, and personalized recommendation pipelines are implemented, but use their fallback behavior until sufficient data is collected and the training command is executed.

2. Deployed model: agent–property KNN matcher

When a manager creates an agent, the system recommends the most suitable listings from the agent profile. It first applies strict eligibility rules for property type, exact territory (city + governorate), and minimum/maximum TND budget. Only eligible properties enter the K-Nearest Neighbors ranking stage.

- Categorical features: property type and territory, encoded with OneHotEncoder.
- Numerical feature: listing price, normalized with StandardScaler.
- Algorithm: scikit-learn NearestNeighbors, brute-force Euclidean distance.
- Query profile: all selected property-type/territory combinations at the agent budget midpoint.
- Output: ranked properties, a normalized match score, and human-readable reasons.

This model is fitted in memory against the current eligible catalog each time recommendations are calculated. It does not require historical labels; its purpose is profile similarity and allocation support, not price prediction.

3. Implemented training pipelines

The backend includes reproducible scikit-learn pipelines. Categorical values are one-hot encoded, numerical values are standardized, and each supervised model uses an 80/20 train–validation split with a fixed random seed. The best candidate is saved with joblib together with metadata, evaluation metrics, feature schema, and a model card.

Model	Algorithms compared	Main inputs	Selection / fallback
Sale & rent price estimator	Gradient Boosting Regressor; Random Forest Regressor	Location, type, area, rooms, floors, year, furnishing and property features	Best R^2; fallback: median price/m^2 from comparable listings
Lead conversion predictor	Gradient Boosting Classifier; Random Forest Classifier	Request type, budget, location, urgency, engagement, appointments and conversations	Best ROC-AUC; fallback: rule-based conversion heuristic
Client property recommender	Content-based KNN (cosine); Truncated SVD collaborative filtering	Property attributes plus weighted views, favorites, messages and appointments	Best hit-rate@5; fallback: popularity-based ranking

Price estimation

Separate sale and rent regressors are supported. Gradient Boosting uses 200 estimators, depth 5 and learning rate 0.1; Random Forest uses 300 trees, depth 15 and minimum leaf size 3. Evaluation includes R^2 and MAE, with residuals retained for indicative prediction intervals. Estimates are advisory and never overwrite an agent-entered price.

Lead conversion prediction

The binary target represents conversion within the configured time window through an appointment or contract. Gradient Boosting (200 estimators) is compared with a class-balanced Random Forest (300 trees). The probability is shown as an experimental agent aid; it does not replace the existing lead score and must not automatically reject a client.

Personalized recommendations

The recommender compares content-based cosine KNN with collaborative Truncated SVD. Interactions form a user–property matrix; leave-one-out evaluation measures precision@5, recall@5, hit-rate@5, NDCG and catalog coverage. Until interaction volume is sufficient, the system returns popularity-based recommendations.

4. Other uses of AI and intelligent automation

Lead scoring and agent matching

- A transparent 0–100 lead score measures budget clarity, location specificity, property criteria, urgency, activity and engagement.
- Agent matching combines property-type, territory, budget, availability and experience scores to rank agents for a client request.
- Scores can be recalculated as messages, saved searches, favorites and appointments change.

AI assistant

The website assistant detects intents, retrieves platform knowledge and proposes role-specific actions such as browsing properties, opening messages, creating appointments or viewing analytics. Its base implementation is deterministic and rule-based. An external OpenAI-compatible generative layer is optional; without an API key, the assistant continues with safe predefined responses and does not fabricate transactions.

Investment score

Investment analysis is not a trained ML model. It is a deterministic weighted formula using market demand (30%), affordability (25%), rental yield (20%), price growth (15%) and liquidity (10%). It provides an indicative score, risks and recommendation using market-area data.

Data and responsible use

- Property data includes type, Tunisian location, TND price, surface, rooms, coordinates, images and features.
- Interaction events include views, favorites, saved searches, messages, appointments and completed transactions; message content and credentials are excluded.
- Every model has a fallback for missing artifacts, insufficient rows, incompatible schemas or inference errors.
- Predictions are advisory, include limitations, and retain human review by agents and managers.

Summary: RE/MAX Smile AI uses KNN directly for agent–property allocation and provides complete training infrastructure for pricing, conversion prediction and personalization. The system deliberately separates true trained ML from deterministic scoring, ensuring that the platform remains explainable and reliable while more real interaction and transaction data is collected.